

Jacob D. Ruth

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Key Projects

Keeper | Unity, C# | 8 person team | Capstone Project

On Keeper, I worked on character animation integration, save and load functionality, and an interactive in-game journal. I iterated on existing systems and managers and helped in fixing bugs and other stability issues. I collaborated with designers and artists over a two semester development cycle to deliver a completed educational game.

Procedural Animation: Spider | Unity, C# | Solo Project

For this project, I implemented a procedural animation system using inverse kinematics, raycasting, and quaternion-based bone rotations by developing a scalable system that dynamically calculates valid surface positions and orientations for each leg and foot. Then the system would animate the leg motion and placement in response to character input.

Pentaflux | Unity, C# | 4 person team | Ludum Dare 49

During a game jam, I designed the player controller and a dynamic tile system that handled spawning, despawning, flipping, and sliding tiles. I created the system to have weighted tile selection to allow designers to control spawn probabilities for enemies, pickups, and environmental tiles. I collaborated with a small team to build a playable game within a 48 hour time constraint.

Relevant Proficiencies

Programming Languages:

C#, C++, C, GDScript, Python, HTML5, JavaScript, Kotlin, Java, MySQL

Game Engines & Tools:

Unity, Unreal Engine, Godot
Visual Studios, Android Studios, Photoshop, 3DS Max, Blender

Platforms:

Windows, Linux, Android, Web

Work Experience

Contractor | Website Development, Burlington, VT | Nov 2018 - Present

I designed and maintained redpoppycakery.net, a customer facing website for a local bakery. I collaborated directly with the owner to implement features and changes as business needs evolved over time. I have been managing ongoing site updates, content changes, and usability improvements including a custom photo gallery and contact form.

Manager | Vivid Coffee, Burlington VT | Nov 2020 - Present

My role as a manager included overseeing daily café operations, ensuring smooth workflows, consistent service quality, and adherence to established procedures. I would train baristas, setting performance expectations. I ordered and maintained inventory to support uninterrupted daily operations. I handled cash deposits with accuracy and accountability.

Shift Lead | New Moon Cafe, Burlington VT | Apr 2017 - Nov 2020

As a Shift Lead, I led staff during shifts by assigning tasks to maintain efficient operations. I also helped train new hires. I was responsible for handling cash and deposits. Additionally, I assisted management with inventory tracking and ordering.

Associate Trainer | Panera Bread, Concord NH | Jul 2014 - Nov 2015 &

Associate Trainer | Panera Bread, Burlington VT | Jul 2016 - Aug 2017

As an associate trainer, I was certified to train new employees in any of the entry-level positions. I had excellent customer service, quick speed of service, and was awarded rookie of the year, at both locations.

Tutor | NHTI Concord's Community College | Sep 2014 - May 2015

While a senior at NHTI I tutored students in the Animation and Graphic Game Programming major, reinforcing core game programming and development concepts by assisting students to understand foundational programming topics.

Education

Champlain College | Bachelor's Degree | Dec 2020

Major: Game Programming

NHTI Concord's Community College | Associate's Degree | May 2015

Major: Animation and Graphic Game Programming